Grade 5 General Science

Solved Exercises

Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: Structure and Function-Human Body System

Unit 1: Structure and Function-Human Body System. The human body is made of organ systems that work together to keep us alive and healthy. The respiratory system brings oxygen into the body, the circulatory system carries oxygen and nutrients to every part, and the nervous system controls everything through the brain, spinal cord and nerves.

1.1 Multiple choice questions.

(i) The organ that pumps blood throughout the body is:

- Lung
- ✓ Heart
- Stomach
- Brain

(ii) The part of the body that controls thinking and movement:

- Heart
- Lungs
- ✓ Brain
- Intestine

(iii) The body system made of nerves, brain, and spinal cord is called:

- Digestive system
- Circulatory system
- Respiratory system

✓ Nervous system

(iv) The tiny blood vessels that connect arteries and veins:

✓ Capillaries

Alveoli

Nerves

Bronchi

(v) The largest artery in the human body is:

Vena cava

Pulmonary vein

✓ Aorta

Jugular vein

(vi) The system that brings oxygen to the body:

Nervous system

Circulatory system

Digestive system

✓ Respiratory system

(vii) A healthy habit for keeping the heart strong:

Eating oily junk food

Skipping sleep

✓ Doing daily exercise

Sitting all day

(viii) The tiny air sacs inside the lungs are called:

Diaphragm

Pharynx

Inhalation

✓ Alveoli

(ix) The control center of the human body is:

Lungs

Heart

✓ Brain

Kidneys

(x) There are two spongy organs located in the chest:

Heart

✓ Lungs

Spinal cord

Vessels

1.2 Fill in the blanks.

(i) The main parts of the nervous system are the ____, ____, and ____.

brain, spinal cord, and nerves

(ii) The exchange of oxygen and carbon dioxide takes place in the lungs inside tiny sacs called ____.

alveoli

(iii) The three types of blood vessels are ____, ____, and ____.

arteries, veins, and capillaries

(vi) Air enters the body through the ____.

nose

(v) The trachea splits into two ____.

bronchi

1.3 Match column A with column B.

Match: Brain, Spinal cord, Lungs, Alveoli, Vertebrae.

Brain: Controls thoughts, memory and body movements. Spinal cord: Carries messages between brain and body. Lungs: Take in oxygen from the air and send it to the blood. Alveoli: Exchanges oxygen and carbon dioxide with blood. Vertebrae: Protects spinal cord inside the back bone.

1.4 Write short answers of the following questions.

(i) Name the Organ that pumps blood in the body.

The heart pumps blood in the body. It is a strong muscular organ about the size of a fist, and it works all the time, even when we are sleeping, beating around 100,000 times a day.

(ii) What are three main parts of the nervous system?

The three main parts of the nervous system are the brain, the spinal cord and the nerves. Together they control everything that happens in the body.

(iii) Name the system that helps in breathing.

The respiratory system helps in breathing. It includes the nose, pharynx, larynx, trachea, bronchi, lungs, alveoli and diaphragm,

and it brings oxygen in and removes carbon dioxide.

(iv) Name the largest artery in the human body.
The largest artery in the human body is the aorta. It carries oxygen-rich blood from the heart to the rest of the body.

(v) Name the blood vessel that connects arteries and veins.
Capillaries are the blood vessels that connect arteries and veins. They are very small tubes that help in the exchange of oxygen, food and waste between the blood and the body cells.

(vi) Suggest one way to keep the heart healthy.
One way to keep the heart healthy is to do daily exercise. Eating well and resting enough also help to keep the heart and the body systems working properly.

1.5 Write detailed answers of the following questions.
(i) Describe the role of the heart, blood and blood vessels in the human circulatory system.
The heart is the pump of the circulatory system: a strong muscular organ with four chambers that beats day and night to send blood around the body. The blood vessels are the tubes that carry the blood: arteries carry blood away from the heart, veins bring blood back to the heart, and tiny capillaries join them and exchange oxygen, food and waste with the body cells. The blood itself is a red liquid made of red blood cells, white blood cells, platelets and plasma; it carries oxygen, food and water to all parts of the body and brings back waste like carbon dioxide. Together the three parts

work like a delivery and cleaning system that keeps us alive, healthy and active.

(ii) Explain how the respiratory system helps in the exchange of oxygen and carbon dioxide in the lungs.

When we breathe in, air enters through the nose, where tiny hair and mucus filter dust and germs and warm and moisten it, then passes through the pharynx and the trachea, which splits into two bronchi, one leading to each lung. Inside the lungs the bronchi divide into smaller tubes called bronchioles, which end in tiny air sacs called alveoli. In the alveoli, oxygen from the air enters the blood and carbon dioxide leaves the blood; this is called gaseous exchange. The oxygen-rich blood is then carried to the body, while the carbon dioxide is breathed out through the same path. The diaphragm, a muscle below the lungs, helps to take air in and push it out.

(iii) Describe the structure and function of the nervous system, including the role of the brain, spinal cord and nerves.

The nervous system is the control center of the body. The brain is its main part: a soft organ protected by the skull that helps us think, learn, remember things and make decisions, and controls our movements, speech and senses like seeing, hearing, smelling and touching. It even controls actions we do not think about, like breathing, digestion and heartbeat, and works all the time, even when we sleep. The spinal cord is a long tube-like structure made of nerves that starts from the brain and runs down the middle of the back inside the backbone; it is protected by the bones of the spine,

called vertebrae, and carries messages between the brain and the rest of the body, for example telling the hand to move away from something hot. The nerves are like tiny wires that carry messages between the brain, the spinal cord and the rest of the body: sensory nerves carry messages from body parts to the brain and help us feel heat, cold, pain and touch, motor nerves carry messages from the brain to the muscles to move our arms, legs and other parts, and mixed nerves do both jobs. Together the three parts keep the body active and connected.

1.6 Constructed response question.

What would happen to the people if there were no oxygen in the atmosphere? Explain how it would affect life on Earth.

If there were no oxygen in the atmosphere, people could not breathe and could not live, because we need oxygen to live and to make energy for our bodies. The respiratory system would have nothing to take in, so oxygen could not reach the blood and the body cells, and the circulatory system could not deliver it. Without oxygen, life on Earth as we know it would end: humans and other animals that breathe oxygen would die, and the systems of the body that depend on oxygen for energy would stop working.

1.7 PROJECT

Project: Fill a syringe with coloured water and push it through a straw to show how the heart pumps blood through blood vessels.

Teacher's note: the syringe is the heart, the coloured water is blood, and the straw is a blood vessel. Pushing the plunger shows how the heart pumps blood through vessels. This is a hands-on classroom

task, not a written answer.

Unit 2: Microorganisms and Diseases

Unit 2: Microorganisms and Diseases. Microorganisms are tiny living things, too small to be seen with the naked eye, found in air, water, soil and even inside our body. The three main types are bacteria, viruses and fungi: some are helpful, making food and medicines and cleaning the environment, while others are harmful and cause diseases.

2.1 Multiple choice questions.

(i) The group that does not include microorganisms is:

- ✓ Sponges
- Viruses
- Fungi
- Bacteria

(ii) The instrument that helps to see microorganisms is:

- Telescope
- Goggles
- Binoculars
- ✓ Microscope

(iii) Microorganism used in making yogurt is:

- Yeast
- Virus
- ✓ Bacteria
- Unicellar algae

(iv) Type of microorganism that needs a living host to multiply is called:

- Fungus
- ✓ Virus
- Bacteria
- Nostoc

(v) A disease caused by bacteria is:

- Polio
- Measles
- ✓ Typhoid
- COVID-19

(vi) Ringworm in humans is caused by:

- Virus
- Bacteria
- ✓ Fungi
- Algae

(vii) Action that helps in preventing contagious diseases:

- Sharing towels
- Not washing hands
- ✓ Covering mouth when sneezing
- Drinking cold drinks

(viii) Example of a non-contagious disease is:

- Flu

✓ Asthma

Polio

Chickenpox

(ix) Correct example of a helpful microorganism:

Bacteria causing food poisoning

✓ Yeast used in bread making

Virus spreading polio

Fungi causing ringworm

(x) Main role of decomposer microorganisms in nature is:

Spreading diseases

Producing oxygen

✓ Breaking down dead matter

Making clothes clean

2.2 Fill in the blanks.

(i) Microorganisms are so small that they can only be seen with a ____.

microscope

(ii) The three main types of microorganisms are ____, ____, ____.

bacteria, viruses, and fungi

(iii) A virus must enter a ____ to become active.

host cell

(iv) Coccus bacteria are ____ shaped.

round

(v) Yeast is a type of fungus which is used to make ____ and ____.
bread and cakes

2.3 Match column A with column B.
Match: Bacteria, Viruses, Fungi, Alveoli, Antonie Van Leeuwenhoek.
Bacteria: Cause diseases like pneumonia, tuberculosis. Viruses: Need a host cell to reproduce. Fungi: Grow on dead matter; some used in baking. Alveoli: Tiny air sacs for gas exchange in humans. Antonie Van Leeuwenhoek: First observed microorganisms in 1683.

2.4 Write short answers of the following questions.
(i) What are microorganisms?
Microorganisms are tiny living things that are too small to be seen by the naked eye, so we need a microscope to see them. They are found everywhere, in air, water, soil and even inside our body, and the three main types are bacteria, viruses and fungi.

(ii) Name any two diseases caused by viruses.
Two diseases caused by viruses are flu and measles. Other examples from the unit are polio, chickenpox, dengue and COVID-19.

(iii) How do bacteria help in food digestion?
Some bacteria in our stomach help us digest food properly. These helpful bacteria live in the gut and help the body break down and use the food we eat, while harmful bacteria instead cause illness.

(iv) What is the difference between contagious and non-contagious diseases?

Contagious diseases spread from one person to another, are caused by harmful microorganisms, and spread through dirty water, touch and contact; they can be prevented by hygiene and isolation. Non-contagious diseases do not spread from person to person or through contact; they are caused by poor diet, genetics or other reasons and can be prevented by a healthy lifestyle.

(v) List two ways to prevent the spread of infectious diseases.

Two ways to prevent the spread of infectious diseases are washing hands regularly with soap and water, and covering the mouth while sneezing or coughing. Eating clean food and avoiding contact with sick people also help.

(vi) What is the role of fungi in nature?

Fungi grow on dead and decaying matter and help in breaking down waste in nature. By decomposing dead plants and animals they keep the environment clean and make the soil rich, and some fungi, like yeast, are also useful in baking.

2.5 Write detailed answers of the following questions.

(i) What are microorganisms? Name the three main types and give one example of each.

Microorganisms are tiny living things too small to be seen with the naked eye, found in air, water, soil and inside our bodies; we need a microscope to see them. The three main types are bacteria, for

example the bacteria that cause pneumonia and tuberculosis; viruses, for example the influenza virus that causes flu; and fungi, for example yeast, which is used in making bread, or mushrooms, which are also fungi.

(ii) How do harmful microorganisms spread from one person to another? Write any three ways to stop their spread.
Harmful microorganisms spread from one person to another through the air, water, food, dirty hands and contact with infected people or objects. When a sick person sneezes or coughs, tiny droplets containing germs mix with the air and another person breathing that air can get sick; drinking unclean water or eating spoiled food, touching dirty surfaces, and shaking hands, sharing utensils or towels and touching doorknobs also spread germs. Three ways to stop their spread are washing hands regularly with soap and water before eating and after coughing or sneezing, wearing a mask in crowded places so germs do not spread through the air, and getting vaccinated, for example polio drops for children and COVID-19 vaccines.

(iii) Explain the differences between contagious and non-contagious diseases? Give two examples of each.
Contagious diseases spread from one person to another, are caused by harmful microorganisms, spread through dirty water, touch and contact, and can be prevented by hygiene and isolation; examples are flu, measles and COVID-19. Non-contagious diseases do not spread from person to person or through contact, are caused by poor diet, genetics or other reasons, and can be prevented by a

healthy lifestyle; examples are diabetes, cancer and asthma.

(iv) How are microorganisms helpful to us? Write about their role in food, medicine, and cleaning the environment.

In food, microorganisms make bread, yogurt and cheese: yeast, a fungus, is used to make bread, cakes and other baked foods, and bacteria are used to make yogurt, cheese and vinegar. In medicine, certain bacteria are used in making medicines such as antibiotics that fight harmful bacteria, and scientists study viruses to create vaccines and medicines that protect us from viral diseases. In cleaning the environment, microorganisms act as decomposers: they break down the dead bodies of living things, which keeps the environment clean and makes the soil rich, and fungi growing on dead and decaying matter help break down waste in nature.

2.6 Sort the following into their respective groups.

Sort into Fungi, Bacteria and Virus: COVID-19, ringworm, polio, tetanus, leprosy, AIDS, athlete's foot and smut.

Fungi: ringworm, athlete's foot, smut. Bacteria: tetanus, leprosy. Virus: COVID-19, polio, AIDS.

2.7 Project

Project: Leave one damp cloth in sunlight and another in a dark place. Compare smell or mold after a few days. Teacher's note: Sunlight kills or slows down microorganisms, showing the importance of hygiene. This is a hands-on classroom task, not a written answer.

Unit 3: Ecosystem

Unit 3: Ecosystem. An ecosystem is a place where living things like plants and animals interact with non-living things such as air, water and soil. Producers make food, consumers eat plants or other animals, and decomposers break down dead things and return nutrients to the soil, so energy flows from the Sun through food chains and food webs.

3.1 Multiple choice questions.

(i) Producers in an ecosystem:

- ✓ Make food using sunlight
- Eat other animals
- Live in water only
- Use energy from other animals

(ii) Decomposers in nature:

- Hunt prey for energy
- Grow tall like trees
- ✓ Break down dead organisms
- Store energy from the Sun

(iii) Main energy source for all life on Earth:

- Wind
- Moon
- ✓ Sun
- Water

(iv) Herbivores are animals that:

- Eat meat only
- Eat both plants and animals
- ✓ Eat plants only
- Use sunlight to grow food

(v) Energy at higher levels in a food chain:

- Increase each step
- Remains equal
- ✓ Drops at each level
- Disappears completely

(vi) Smoke from factories and vehicles:

- Noise pollution
- Land pollution
- ✓ Air pollution
- Light pollution

(vii) Plastic bags and glass bottles:

- Break down quickly in soil
- Add nutrients to plants
- ✓ Remain in nature for years
- Help composting

(viii) Food Web:

- ✓ Connects many food chains
- Grows in oceans

Includes only plants

Works without animals

(ix) Removal of forests for buildings:

Urbanization

Mining

✓ Deforestation

Composting

(x) Predators:

Depend on sunlight for food

Escape from larger animals

✓ Hunt and eat other animals

Clean waste from the environment

3.2 Fill in the blanks.

(i) Green plants make their own food through a process of ____.

photosynthesis

(ii) Animals that eat only plant are called ____.

herbivores

(iii) Animals like lions and tigers that eat only animals are called ____.

carnivores

(iv) Living things like fungi and bacteria that break down dead plants and animals are called ____.

decomposers

(v) In a food chain, the main source of energy for all living things is the ____.
Sun

3.3 Match column A with column B.
Match: Producers, Herbivores, Carnivores, Omnivores, Decomposers.
Producers: Green plants that make food using sunlight. Herbivores: Animals that eat only plants (cow and rabbit). Carnivores: Animals that eat only other animals (lion and eagle). Omnivores: Animals that eat both plants and animals (bear and human). Decomposers: Break down dead plants and animals (fungi and bacteria).

3.4 Write short answers of the following questions.
(i) What is a producer?
A producer is a living thing, like a green plant, that makes its own food using sunlight, air and water through photosynthesis. Producers make food not only for themselves but also for other living things, and without them animals would have nothing to eat.

(ii) Name one example of a decomposer.
One example of a decomposer is fungi. Bacteria are also decomposers; they break down dead plants and animals and return nutrients to the soil.

(iii) What type of consumer is a lion?

A lion is a carnivore. Carnivores are animals that eat only other animals; they hunt or eat meat to get energy, and the unit gives lion, tiger and eagle as examples.

(iv) Which gas is released by burning fossil fuels?

Burning fossil fuels releases harmful gases into the air and adds carbon dioxide to the atmosphere. The carbon dioxide traps heat and causes global warming.

(v) Define non-biodegradable materials. Give an example.

Non-biodegradable materials are things that do not break down easily in nature; they stay in the environment for many years, cannot be decomposed by bacteria or fungi, and can cause pollution. An example is plastic bags; glass bottles, aluminum cans and rubber are other examples.

3.5 Write detailed answers of the following questions.

(i) Explain how a food chain works using an example.

A food chain shows how energy moves from one living thing to another in an ecosystem, starting with the Sun, which gives energy to plants, the producers. The unit's example is: Sunlight to Grass, to Grasshopper, to Frog, to Snake, to Eagle. Grass uses sunlight to make food; the grasshopper eats the grass and is a herbivore and primary consumer; the frog eats the grasshopper and is a secondary consumer; the snake eats the frog; and the eagle eats the snake and is the apex predator. Each step is called a trophic level, and when these living things die, decomposers break them down and return nutrients to the soil, helping the food chain to

start again.

(ii) Describe the impact of non-biodegradable materials on the environment.

Non-biodegradable materials like plastics, glass and metal stay in the environment for a very long time without breaking down; plastic bags can take over 500 years to break down in the soil. When thrown into land, water or oceans they cause severe pollution: animals may eat plastic, which can make them sick or even kill them, they block drains and cause flooding, and they pollute the soil and water. Burning them can release harmful gases into the air. To protect the environment we must reduce the use of non-biodegradable items and reuse or recycle them whenever possible.

(iii) Describe the effects of burning fossil fuels on the environment and living things.

When fossil fuels like coal, oil and gas are burned for energy, they release harmful gases into the air, causing air pollution that can lead to breathing problems and asthma in people and harm birds and animals. Burning fossil fuels also adds carbon dioxide to the atmosphere, which traps heat and causes global warming, leading to climate change, melting ice, rising sea levels and stronger storms. It harms animals, plants and the balance of nature, which is why the unit says we should use natural energy sources like solar and wind power whenever possible.

3.6 Classify the animals into herbivores, carnivores and omnivores

and write their names in their respective columns.

Classify: Cow, Lion, Wolf, Bear, Deer, Rabbit, Fish, Falcon, Tiger.

Herbivores: Cow, Deer, Rabbit. Carnivores: Lion, Wolf, Falcon, Tiger. Omnivores: Bear, Fish. (Note: the unit names deer as a primary consumer that eats plants, wolf as a predator, and bear as an omnivore; fish is placed in Omnivores as a model answer — see the flags chapter, because the unit lists fish only as an example of prey and does not state its diet.)

3.7 Constructed response question.

How does energy flow through a food chain and why is some energy lost at each level?

Energy flows through a food chain from the Sun to producers, then from one living thing to another: plants, the producers, use sunlight to make their food; when herbivores eat plants they get some of that energy; and when carnivores eat the herbivores they also get energy. Each step is called a trophic level. As energy moves from one level to the next, some energy is lost, and energy decreases at each trophic level, because most of it is used by the living things for activities like moving, growing and keeping themselves warm, and some is lost as heat, as shown in the energy pyramid.

3.8 PROJECT

Project: Water one plant with clean water and another with slightly soapy water. Compare growth after a week. Teacher's note: Demonstrates how pollution affects living things in an ecosystem. This is a hands-on classroom task, not a written answer.

Unit 4: Physical and Chemical Changes of Matter

Unit 4: Physical and Chemical Changes of Matter. Matter is anything that has mass and occupies space, and it exists in three states: solid, liquid and gas. In a physical change only the state, shape, colour or taste changes and no new substance is formed, while in a chemical change new substances with different properties are formed by a chemical reaction.

4.1 Multiple choice questions.

(i) Which is a physical change?

- Burning of wood
- Rusting of iron
- ✓ Melting of ice
- Decaying

(ii) Change forming new substance:

- Physical
- ✓ Chemical
- Temporary
- Reversible

(iii) Heat is not required for:

- Boiling
- Burning
- Evaporation
- ✓ Condensation

(iv) Molecules move apart, distance increases, state results:

Solid

Liquid

✓ Gas

Sublimation

(v) Chemical reaction of material/substance with oxygen is called:

✓ Burning

Rusting

Cooking

Decaying

(vi) Decaying/food spoilage caused by:

Bacteria

Fungi

✓ Both bacteria and fungi

Amoeba

4.2 Fill in the blanks.

(i) Melting/freezing are examples of ____ type of change.

physical

(ii) Iron door painted to avoid ____.

rusting

(iii) Permanent changes result in ____ change.

chemical

(iv) Dissolution of sugar in water is example of ____.

physical change

(v) Burning also produces light and ____.

heat

4.3 Match column A with column B.

Match: Matter, Melting, Freezing, Boiling, Condensation, Rusting.

Matter: Anything that has mass and occupy space. Melting: Conversion of solid into liquid. Freezing: Conversion of liquid into solid. Boiling: Conversion of liquid into vapour upon heating. Condensation: Conversion of vapours into liquid. Rusting: Reaction of iron with oxygen.

4.4 Write short answers of the following questions.

(i) Give two examples of physical change from daily life.

Two examples of physical change from daily life are the melting of ice cubes into liquid water when heated, and the dissolving of sugar in water, which can be recovered again by evaporating the water. Other examples are squeezing a tin can to change its shape and the change in colour and taste when a drink powder is dissolved in water.

(ii) Define burning.

Burning, or combustion, is the chemical reaction of a material in the presence of oxygen to produce heat and light. For example, when wood is burnt it changes into coal or ash, carbon dioxide and water

vapours, and smoke is produced and heat is generated.

(iii) Define condensation.

Condensation is the change of state of matter from gaseous to liquid. It is the reverse process of evaporation: heat is removed from the gaseous state, the vapour molecules strike a colder surface, lose their energy and come closer together, and are converted into liquid, as in water droplets on a cold glass or on a car window in winter.

(iv) When wood burns, which substances are produced?

When wood burns, it changes into coal or ash, carbon dioxide and water vapours. Smoke is also produced and heat is generated during the burning.

(v) Can chemical changes be reversed and why?

Chemical changes are mostly irreversible, because new substances with different properties are formed and the molecular arrangement changes, with old bonds breaking and new bonds forming. For example, coal or ash cannot be converted back into wood, and a rusted iron door cannot be turned back into unrusted iron.

4.5 Write detailed answers of the following questions.

(i) Compare physical and chemical changes.

A physical change is reversible, for example solid ice can be converted back into water; no chemical reaction occurs, no new substance is formed, and the molecular arrangement remains the

same, as in evaporation, condensation, squeezing and dissolving. A chemical change is mostly irreversible, for example coal or ash cannot be converted back into wood; a chemical reaction takes place, a new substance is formed, and the molecular arrangement changes as old bonds break and new bonds are formed, as in decaying, burning and rusting.

(ii) Differentiate between boiling and condensation.

Boiling is the change of state of matter from liquid to gaseous state in the form of vapours upon heating: when a liquid is heated, its particles move faster and farther apart, intermolecular forces become weaker, and the liquid is converted into vapours. Condensation is the opposite change, from gaseous to liquid: it is the reverse process of evaporation, heat is removed from the gaseous state and given to the atmosphere, and when vapour molecules come into contact with a colder surface they lose energy, come closer and are converted into liquid.

(iii) Write a note on (a) Rusting (b) Burning.

(a) Rusting: The chemical reaction of iron with oxygen and moisture from the air is called rusting. The reddish-brown substance or layer developed on iron doors, locks and chains is due to rusting; when metal rusts it gets damaged and the process cannot be reversed. Iron doors are painted to prevent rusting. (b) Burning: The chemical reaction of a material in the presence of oxygen to produce heat and light is called combustion or burning. When wood is burnt it changes into coal or ash, carbon dioxide and water vapours, and the burning of fuel or natural gas also involves reaction with oxygen

and production of carbon dioxide, water and heat, with smoke produced and heat generated.

(iv) Explain the process of evaporation and give examples.
Evaporation is the change of state of matter from liquid into gaseous state or vapour, without heating. Liquid particles are in constant motion, and when some particles with high energy come to the surface they gain enough energy from the surroundings, leave the surface and evaporate. Water can evaporate at any temperature; it can take place from the leaves of plants, seas, rivers and canals, and although actual heating is not required, it is still temperature dependent and occurs faster at higher temperatures, which is why wet clothes and wet surfaces dry, and a shirt hung in sunlight dries earlier than one hung in a room. Evaporation is important for supporting life on Earth and maintaining the water cycle, since rainfall depends on clouds formed by evaporation.

(v) Name the processes involved in chemical changes and write briefly about them.
The processes involved in chemical changes are decaying, burning and rusting. Decaying is the natural breakdown or decomposition of plants, animals and food by microorganisms, mainly bacteria and fungi, which is why dead bodies disappear from the ground and damaged fruit rots, changing colour, becoming softer and giving off a distinct odour. Burning is the chemical reaction of a material with oxygen to produce heat and light, changing wood into coal or ash, carbon dioxide and water vapours. Rusting is the chemical reaction

of iron with oxygen and moisture from the air, forming a reddish-brown layer that damages the metal and cannot be reversed.

4.6 Constructed response questions.

(i) Will iron get rust if there is no moisture?

No, iron will not get rust if there is no moisture, because rusting is defined as the chemical reaction of iron with oxygen and moisture from the air. Without moisture, one of the two things needed for rusting is missing, so the reddish-brown rust layer cannot form.

(ii) Why does the dead body of an animal or plant disappear from the ground after some days?

The dead body disappears because of the decaying action of microorganisms. Decaying is the natural breakdown or decomposition of plants, animals and food by microorganisms, mainly certain bacteria and fungi, which break the body down until it is gone from the ground.

4.7 PROJECT

Project: Mix vinegar with water to show a physical change, because no new substance is formed, it is just dilution; then mix vinegar with baking soda to show a chemical change, because bubbles of carbon dioxide gas are produced. Teacher's note: the same substance, vinegar, takes part in both changes. This is a hands-on classroom task, not a written answer.

Unit 5: Light and Sound

Unit 5: Light and Sound. Light and sound are both forms of energy. Light is the phenomenon through which we see our surroundings, and it can travel even in a vacuum, while sound is produced by vibrations and needs a medium for its propagation.

5.1 Multiple choice questions.

(i) Largest source of light on Earth:

- Moon
- ✓ Sun
- Electricity
- Mars

(ii) Sound waves travel fastest through:

- Air
- Water
- ✓ Steel
- Vacuum

(iii) Opaque object:

- Clear glass
- Plastic sheet
- Water
- ✓ Wooden door

(iv) Objects across which we can see clearly:

Translucent

Opaque

✓ Transparent

Absorbent

(v) Objects emitting own light:

✓ Luminous

Non-luminous

Burning

Dark

(vi) Bulb falls under source of light:

Natural

✓ Artificial

Both

Non-luminous

5.2 Fill in the blanks.

(i) Light does not need ____ for propagation.

medium

(ii) Sound cannot travel through ____.

vacuum

(iii) Louder sounds have ____ intensity.

higher

(iv) Objects not emitting own light are called ____.

non-luminous

(v) Undesired sounds are called ____.

noise

5.3 Match column A with column B.

Match: Light, Artificial, Non-Luminous, Loss of hearing, Vibrations.

Light: Form of energy. Artificial: Bulb. Non-Luminous: Book. Loss of hearing: Noise. Vibrations: Sound.

5.4 Write short answers of the following questions.

(i) Differentiate between natural and artificial sources of light.

Natural sources of light emit light without the involvement of human beings, for example the Sun, stars and the Moon. Artificial sources of light emit light due to the involvement of human beings, for example bulbs and lasers.

(ii) Does sound need medium for its propagation?

Yes, sound needs a medium for its propagation, unlike light which can travel even in a vacuum. The medium can be a solid, liquid or gas, and sound cannot travel where there is no medium, which is why sounds from outer space do not reach the Earth.

(iii) Give three examples of non-luminous objects.

Three examples of non-luminous objects are a book, a wall and a man. Such objects do not have their own light to emit; we see them when light gets reflected from them.

(iv) How is sound produced?

Sound is produced by the vibrations of an object. When a body vibrates it pushes the particles of the air, the air particles get compressed, and the continuous vibration of the source causes the air particles to vibrate, so sound waves start travelling in the air.

(v) Define translucent objects. Give two examples.

Translucent objects are those which allow some of the light to pass through them, so we cannot see clearly through them. Two examples are frosted glass and coloured plastics; wax papers are another example from the unit.

5.5 Write detailed answers of the following questions.

(i) Explain the conduction mechanism of sound waves.

Sound is produced by a vibrating body. Whenever a body vibrates, it pushes the particles of the air, and in that region the air particles get compressed. The continuous vibration of the source causes the air particles to vibrate, and the sound waves start travelling in the air as well, with regions of increased pressure and decreased pressure moving in the direction of propagation. Sound travels in the form of waves and needs a medium: the speed is fastest in solids, where particles are arranged very close to each other, slower in liquids, where there is more gap between particles, and slowest in air or gases, where the space between particles is larger. The unit gives approximate speeds of 5940 metres per second in steel, 1496 metres per second in water and 343 metres per second in air.

(ii) Write about the harmful effects of noise on human health.

Noise is not only unwanted and unpleasant; persistent exposure to noise can cause severe health problems. Prolonged exposure can damage hearing and create the perception of ringing and buzzing in the ears even without noise; a sleeping man gets disturbed; it is difficult to study and concentrate in constant noise; the productivity of a worker in the office or work site reduces; noise can cause high blood pressure and increased risk of heart diseases; continuous noise can cause stress and anxiety; digestive issues and diabetes problems can also result from high exposure; and in a noisy area we cannot understand the talks of one another.

(iii) Describe the types of objects on the basis of transmission of light.

On the basis of the transmission of light, objects are classified as transparent, translucent and opaque. Transparent objects allow all the light to pass through them, and we can clearly see through them, for example air, clear glass and camera lenses. Translucent objects allow some of the light to pass through them, so we cannot see clearly through them, for example frosted glass, coloured plastics and wax papers. Opaque objects do not allow light to pass through them, and we cannot see through them, for example a book, a wall and wood.

(iv) How can noise pollution be controlled?

Noise pollution can be controlled by steps taken by humans, since it is caused as well as controlled by humans. Use of horns should be avoided and banned in residential areas, especially near hospitals

and schools; use of loud speakers should be discouraged and banned in public places; sound-proof houses should be built or heavy curtains used to block noise from outside; seminars should be conducted for awareness about the effects of noise pollution; factories should be set up away from residential areas; plantation drives should be encouraged and trees grown around roadsides and streets, as they help block noise; old vehicles creating huge noise must be banned and electric car usage promoted; and heavy machinery at construction sites near residential areas must not be allowed during the night time.

5.6 Constructed response questions.

(i) The largest source of light for planet Earth is?

The largest source of light for planet Earth is the Sun. It is a natural source of light, and its light reaches the Earth in 8 minutes and 20 seconds.

(ii) What kind of objects are lenses of a camera?

The lenses of a camera are transparent objects. Transparent objects allow all the light to pass through them, and we can clearly see through them; the unit lists air, clear glass and camera lenses as examples.

(iii) Why does a person get irritated in a noisy place?

A person gets irritated in a noisy place because noise is an unwanted and undesirable sound. Persistent noise is unpleasant and disturbs sleep and concentration, causes stress and anxiety, and makes it difficult to understand what others are saying, so

being surrounded by it makes a person irritated.

5.7 PROJECT

Project: Place a glass of water in sunlight and hold a white paper behind it to see the rainbow colours. Teacher's note: Demonstrates refraction; white light splits into many colours. This is a hands-on classroom task, not a written answer.

Unit 6: Electricity and Magnetism

Unit 6: Electricity and Magnetism. Electric current is the flow of charges through a specific path, the electric circuit. A magnet is a metallic element that attracts other metallic elements like iron and nickel, and it always has two poles, North and South, where like poles repel and unlike poles attract.

6.1 Multiple choice questions.

(i) Electric current in metals due to flow of:

- ✓ Electrons
- Protons
- Neutrons
- Atoms

(ii) Part of circuit using current to perform work:

- Switch
- Battery
- Wire
- ✓ Load

(iii) North Pole brought close to South Pole:

- They repel
- ✓ They attract
- Nothing happens
- They will produce sound

(iv) Magnetic material:

- Aluminum
- Plastic
- Glass
- ✓ Iron

(v) Magnetic force when magnet moved away:

- Gets stronger
- ✓ Gets weaker
- Remains same
- Doubles

(vi) Freely suspended bar magnet aligns:

- ✓ North-South
- East-West
- North-West
- South-West

6.2 Fill in the blanks.

(i) Complete path for current is called ____.

electric circuit

(ii) Two ends of magnet are called ____.

poles

(iii) Battery acts as a ____ in an electric circuit.

power source

(iv) Steel is ____ materials as it contains iron.

magnetic

(v) Two North Poles brought together will ____ each other.

repel

6.3 Write short answers of the following questions.

(i) Name two household items which use electricity.

Two household items which use electricity are the washing machine and the refrigerator. The unit also names ovens, laptops, mobile phones and electric vehicles as things that need electric current to perform their function.

(ii) Why is switch used in a circuit?

A switch is used in a circuit to block or open the circuit, that is, the path of charges. When the switch is turned on, the circuit is complete and current flows, as in the toy car that starts moving when its switch is turned on; when it is off, the path is blocked and no current flows.

(iii) What is a magnet?

A magnet is a specific metallic element that has the property to attract other metallic elements, such as iron and nickel. A magnet can be in different shapes, and it consists of two ends called the poles, one called the North Pole and the other the South Pole.

(iv) How can two magnets be made to attract each other?

Two magnets can be made to attract each other by bringing

opposite poles close together, for example the North Pole of one magnet near the South Pole of the other. Like poles repel each other, while unlike poles attract, so bringing two North Poles together will repel, not attract.

6.4 Match column A with column B.

Match: Load, Repulsion, Compass, Switch, Current.

Load: LED bulb. Repulsion: Like poles. Compass: North direction. Switch: Control flow of charges. Current: Flow of charges.

6.5 Write detailed answers of the following questions.

(i) Differentiate between magnetic and non-magnetic materials and give examples of each.

Magnetic materials are those which get attracted towards a magnet, for example iron, steel, nickel and cobalt. Non-magnetic materials are those which are not attracted towards a magnet, for example aluminum, copper, gold, wood and silver. The unit's picture shows steel, cobalt, iron and nickel as magnetic materials, and gold, brass, copper, silver and mercury as non-magnetic materials.

(ii) Explain the components of an electric circuit and the working of each component.

An electric circuit consists of five parts. The power source, which may be a battery or an electric cell, provides the necessary energy that pushes the electrons to flow, so the electric current is produced; it works like the water tank on the roof of a house. The wires are made of conductors and provide a path for current flow

from the source to the load. The switch is used to block or open the circuit, the path of charges. The load may be anything that consumes electric current to perform work, for example an electric bulb, a refrigerator or a television. The resistor is used when we need to limit the current flow in a circuit.

(iii) Write a short note on the structure of an atom.

Matter is made up of tiny particles called atoms, and an atom is further made up of three smallest fundamental particles: electrons, protons and neutrons. Protons and neutrons are present inside the nucleus, while electrons revolve around the nucleus in defined paths called orbits. Electrons have negative charges, protons have positive charges and neutrons are neutral, so the nucleus has a positive charge due to the positive charge of protons. The flow of charges, mainly electrons, through a specific path is called the electric current: in a cell the negative end has an excess of electrons and the positive end has a deficiency, so when the circuit is closed the electrons move from the negative end towards the positive end.

6.6 Constructed response questions.

(i) How will life be impacted if there is no electricity?

If there is no electricity, life would become very difficult, because electricity is making our lives easier and bringing comfort in our day to day activities. Without electric current, the technological era would not be possible: washing machines, ovens, refrigerators, laptops, mobile phones and electric vehicles could not perform their functions, and the gadgets and digital tools we depend on

every day would stop working.

(ii) Can a needle be made into a magnet? If yes, how?

Yes, a needle can be made into a magnet. Take a needle, hold it from one side, and rub it in the same direction from any end of a bar magnet, repeating the rubbing 40 to 50 times. The needle becomes magnetized: if you tie it with a thread and let it move freely, it will come to rest in the North-South direction, just like the needle of a directional compass, which is itself a magnetized piece of metal.

6.7 PROJECT

Project: Rub a balloon on your hair and bring it near small bits of paper. Teacher's note: Static electricity builds charges that attract objects. This is a hands-on classroom task, not a written answer.

Unit 7: Technology in Everyday Life

Unit 7: Technology in Everyday Life. Technology is the use of science and scientific tools to solve problems and make lives better. Using scientific knowledge, we design, create and build things easily, from electric bulbs and street lights to roads, railway tracks and bridges, and we use instruments like the spirit level and safety measures for earthquakes and fires.

7.1 Multiple choice questions.

(i) Why technical models built before actual model?

To make it look beautiful

✓ To test ideas and plan carefully

To save money

To play with it

(ii) Which lead of LED is positive?

Shorter

✓ Longer

Both

Medium

(iii) Spirit level uses what to show evenness?

Light beam

✓ Air bubble

Weight

Waves

(iv) During earthquake, action to take:

Run outside

Stay near window

✓ Drop, cover and hold on

Run inside

(v) During fire, if you see smoke, what do you do?

Hold your breath

Run

✓ Lower yourself

Go back to get your bag

(vi) Technology is making lives:

✓ Easier

Boring

Difficult

Slow

7.2 Fill in the blanks.

(i) Technology is using science to ____ problems.

solve

(ii) Shorter leg of LED is the ____ terminal.

negative

(iii) LED stands for ____.

Light Emitting Diode

(iv) Bubble exactly in center of vial/tube means surface is ____.

level

(v) Measures taken during emergency are called ____.

safety measures

7.3 Write short answers of the following questions.

(i) Define technology and give two examples.

Technology is the use of science and scientific tools to solve problems and make lives better; using scientific knowledge, we design, create and build things easily. Two examples are the electric bulb and the street light system, and roads, railway tracks and bridges are other examples from the unit.

(ii) Why do engineers build technical models?

Engineers build technical models before the actual construction so they can test ideas and plan carefully. A technical model is a smaller, simpler version of the real thing: the design is first made on paper or computers, then the smallest mini-model is made, and the technical model is tested for different conditions, and only after all the necessary tests does the actual construction start.

(iii) What are the three main steps during an earthquake?

The three main steps during an earthquake are drop, cover and hold on: drop to the ground and get on your hands and knees, cover your head and neck with one hand and crawl under a study table or any desk, and hold on to the table or desk and stay

dropped down in this position until the shaking stops.

(iv) What is a spirit level? Write its uses.

A spirit level is a small and simple apparatus used to find out the horizontal evenness of a surface. It is a handy tool, usually made of plastic, having a small tube filled with liquid and an air bubble inside. It is mostly used by masonry workers and carpenters to check whether a surface, like a table or a picture frame, is even: if the bubble comes exactly in the center, the surface is perfectly flat, and if the surface is uneven, the bubble moves towards the higher side. Most spirit levels also have a vertical vial for checking vertical level.

7.4 Match column A with column B.

Match: LED, Spirit level, Technology, Longer leg of LED, Shorter leg of LED.

LED: Light Emitting Diode. Spirit level: Even and Uneven surface. Technology: Making lives easier. Longer leg of LED: Positive terminal. Shorter leg of LED: Negative terminal.

7.5 Write detailed answers of the following questions.

(i) Write safety measures to be adopted during an earthquake.

During an earthquake, the following safety measures should be adopted. If you are inside the room: drop to the ground and get on your hands and knees; cover your head and neck with one hand and crawl under the study table or any desk; hold on to the table or desk; stay dropped down in this position until the shaking stops; and never try to jump from the upper floors and never run during

an earthquake. If you are outside: move to a safe area away from buildings and traffic; drop to the ground at that place until the shaking stops; and avoid staying near power lines. Also, be aware of the earthquake shelter area and earthquake active zones near your area.

(ii) Write safety measures to be adopted during a fire.

During an event of fire, follow these measures. First, activate the fire alarm of the building if available, and if a fire alarm is not present, shout 'Fire, Fire, Fire'. Keep on following the escape plan by looking at the fire exit signs. If there is smoke, lower yourself to the ground and move on. Never use the elevator during a fire. After going outside, call the fire extinguisher team and rescue team immediately. Use the fire extinguisher if the fire is small. If clothes catch fire, do not run: drop to the ground, cover your hands and roll over and over, or use a fire blanket to smother the flames.

(iii) Write a note on spirit level.

A spirit level is a small and simple apparatus used to find out the horizontal evenness of a surface, for example when a chair and table placed over a surface start wobbling because the surface is uneven. It is a handy tool, usually made of plastic, having a small tube with a liquid and an air bubble inside. Upon placing the spirit level over any surface, if the bubble comes exactly in the center, the surface is perfectly flat; in case of unevenness, the bubble will move towards the higher side. This instrument is mostly used by masonry workers and carpenters to check the level of different objects, like tables, pictures and frames, and most spirit levels also

have a vertical vial for checking vertical level.

7.6 Constructed response questions.

(i) How is technology improving our lives?

Technology is improving our lives by using science and scientific tools to solve problems and make lives better. From the electric bulb to the whole system of street lights, from roads to railway tracks and bridges, technology is all around us, and in a vast library containing thousands of books we can find a specific one with just one click on a computer. Using scientific knowledge, we design, create and build things easily, so everyday tasks become easier, faster and more comfortable.

(ii) How can a surface be made perfectly horizontal with the help of a spirit level?

Place the spirit level on the surface and observe where the bubble moves. If the bubble is exactly in the center, the surface is perfectly flat and horizontal. If the bubble moves to one side, the surface is uneven and the bubble moves towards the higher side, so adjust the surface, lowering the higher side or raising the lower side, until the bubble comes exactly to the center; at that point the surface is perfectly horizontal.

7.7 PROJECT

Project: Use coloured paper in red, yellow and green with a flashlight or LED to show how traffic lights control movement.
Teacher's note: Highlights the role of technology in road safety. This is a hands-on classroom task, not a written answer.

Unit 8: Structure of the Earth

Unit 8: Structure of the Earth. The Earth is our home and the only planet in our solar system where life exists. It consists of layers, each with its own structure, features, materials and temperature, and understanding these layers helps us learn more about earthquakes and volcanoes.

8.1 Multiple choice questions.

(i) Outermost layer:

✓ Crust

Mantle

Inner core

Outer core

(ii) Hottest layer:

Crust

Mantle

✓ Inner core

Outer core

(iii) Magma reaching surface is called:

Vent

✓ Lava

Volcano

Fire

(iv) Earthquake due to movement of:

✓ Tectonic plates

Core

Mantle

Outer core

(v) Volcano mountain/hills are:

✓ Cone-Shape

Triangle shape

Square shape

Rectangle shape

(vi) Mantle lies beneath:

Core

✓ Crust

Inner core

Outer core

8.2 Fill in the blanks.

(i) Core consists of ____ parts.

two

(ii) Crater found at top of ____.

volcano

(iii) Earthquake caused by movement of ____.

tectonic plates

(iv) Mantle extends up to depth ____ below Earth surface.

2900 km

(v) Outer core is liquid layer consist of ____ and ____.

iron and nickel

8.3 Match column A with column B.

Match: Volcanoes, Earthquake, Crust, Mantle, Magma chamber.

Volcanoes: Lava. Earthquake: Tectonic plate. Crust: Outermost layer. Mantle: Middle layer. Magma chamber: Molten rock stored.

8.4 Write short answers of the following questions.

(i) Why is Earth special as compared to other planets?

Earth is special because it is the only planet in our solar system where life exists. It provides us land to live on, air to breathe, food to eat and water for drinking and other purposes; about 70% of the Earth's surface is covered with water and 30% is land.

(ii) Which layer of Earth is the thinnest?

The crust is the thinnest layer of the Earth. It is the outermost layer, made of solid rock, with a temperature around 22 degrees Celsius; the oceanic crust is 8 km thick, while the continental crust is 8 km to 70 km thick.

(iii) Why is the interior of the Earth hot?

The interior of the Earth gets hotter with depth, as shown by the temperatures the unit gives for each layer: the crust is around 22 degrees Celsius, the mantle is made of denser hot rocks, some

almost melted, the outer core is extremely hot at about 4000 degrees Celsius, and the inner core is extremely hot at between 5000 and 6000 degrees Celsius. The unit states these temperatures but does not give a cause for the heat, so this answer describes the pattern the book itself gives.

(iv) Define Magma chamber, Vent and Crater.

Magma chamber: A large underground chamber where molten rock is stored. Vent: The passage through which magma travels to reach the surface of the Earth, where it is then called lava. Crater: The opening at the top of the volcano where lava, ash and gases erupt.

8.5 Write detailed answers of the following questions.

(i) Describe the structure of the Earth.

The Earth is divided into three main layers. The crust is the outermost layer, a layer of solid rock and the thinnest layer, with a temperature around 22 degrees Celsius; it has two types, the oceanic crust, 8 km thick and made of iron, oxygen, silicon, magnesium and aluminum, and the continental crust, 8 km to 70 km thick and made of granite, sedimentary rocks and metamorphic rocks, which is the part where we live and where most natural resources are found. The mantle lies beneath the crust; it is the largest layer, mostly solid and made of denser hot rocks, some almost melted, approximately 2900 km thick. The core is the central layer, made of iron and nickel, and it has two parts: the outer core, a layer of liquid iron and nickel starting about 2890 km beneath the surface and going down to about 5150 km, around 2260 km thick, at about 4000 degrees Celsius, whose moving

molten metal creates the Earth's magnetic field; and the inner core, a solid ball of iron and nickel about 2500 km wide and about 1220 km thick, from about 5150 km down to 6371 km, at between 5000 and 6000 degrees Celsius.

(ii) Describe the main features of volcanoes in detail.

A volcano is a cone-shaped mountain or hill that forms when hot and melted rock, lava or ash, comes out from inside the Earth, and volcanoes are usually found in breaks or cracks in the Earth's crust. This melted rock is known as magma while it is still underground, and when magma comes to the surface and cools it becomes solid rock. The main features are: the magma chamber, a large underground chamber where molten rock is stored; the vent, the passage through which magma travels to the surface, where it is called lava; and the crater, at the top of the volcano, where lava, ash and gases erupt. Some volcanoes are under the oceans, where they erupt in the form of new islands.

(iii) How are earthquakes caused?

Earthquakes are the vibrations caused on the surface of the Earth when the tectonic plates move. The surface of the Earth is not in one big piece; it is made up of many large pieces called tectonic plates, which are always moving slowly. Sometimes these plates slip against each other, sometimes one plate slides over another, and sometimes the plates move away from each other, and this movement is called plate tectonics. The edges of the plates can get stuck even though the rest of the plate keeps moving, which creates pressure or strain, and when the pressure becomes too

strong the rocks suddenly break or slip, causing the Earth to shake and creating an earthquake.

8.6 Constructed response questions.

(i) Why is the inner core so hot? Its temperature reaches 6000 degrees Celsius.

The inner core is the very centre of the Earth, a huge metal ball of solid iron and nickel about 2500 km wide, and its temperature is extremely high, between 5000 and 6000 degrees Celsius. The temperature rises with depth through the Earth: the crust is around 22 degrees Celsius, the mantle is made of hot rocks, and the outer core is about 4000 degrees Celsius, so the innermost layer is the hottest of all. The unit states these temperatures but does not explain the cause of the heat, so this answer reports the book's own figures rather than adding a cause the book does not give.

(ii) What steps should be taken during an earthquake?

During an earthquake, drop, cover and hold on: drop to the ground and get on your hands and knees, cover your head and neck with one hand and crawl under a study table or any desk, hold on to the table or desk, and stay dropped down in this position until the shaking stops. Never try to jump from the upper floors and never run during an earthquake. If you are outside, move to a safe area away from buildings and traffic, drop to the ground there until the shaking stops, and avoid staying near power lines. Also be aware of the earthquake shelter area and earthquake active zones near your area.

8.7 PROJECT

Project: Collect samples of igneous, sedimentary and metamorphic rocks and label them. Teacher's note: Helps identify the different types of rock making up the Earth's crust. This is a hands-on classroom task, not a written answer.

Unit 9: Soil

Unit 9: Soil. Soil is a thin layer that covers the surface of the Earth and is extremely important for all living things. It is classified into sandy, silty, clayey and loamy types, and it is composed of minerals, water, air and organic contents.

9.1 Multiple choice questions.

(i) Sandy soil is usually found near:

- Forest
- Swaps
- ✓ Deserts and sea side
- Mountains

(ii) The type of soil, which is best for plants growth:

- Sandy soil
- Clayey soil
- ✓ Loamy soil
- Silty soil

(iii) The soil which holds water for the longest time is:

- Sandy
- ✓ Clayey
- Silty
- Loamy

(iv) The soil which is dark in colour and helps in plant growth is:

Clay

Sand

✓ Humus

Water

(v) Geologist studies about:

Fossils

Water

✓ Earth layer

Air

(vi) Compared to sandy soil, silty soil has:

Large particle

More air spaces

✓ Small particle

No water holding capacity

(vii) The Earth inner core is about:

1500 km wide

4000 km wide

✓ 2500 km wide

5150 km wide

9.2 Fill in the blanks.

(i) Soil is a mixture of ____, water, air and organic matter.

mineral particles

(ii) Clayey soil is ____.

fine and sticky

(iii) Farmers grow crops mostly in ____ soil because it is rich and fertile.
loamy

(iv) The method rock beneath the Earth's surface is called ____.
magma

(v) The temperature of the inner core ranges between ____ °C to ____ °C.
5000 °C to 6000

9.3 Match column A with column B.
Match: Palaeontologists, Sandy soil, Loamy soil, Seismologists, Geologists.
Palaeontologists: Studies fossils. Sandy soil: Sea or deserts. Loamy soil: River, flood. Seismologists: Studies earthquake. Geologists: Studies rocks.

9.4 Write short answers of the following questions.
(i) Why is it important to test the soil before planting?
It is important to test the soil before planting because different soils have different compositions, made of different amounts of sand, clay, silt and organic matter, and the composition decides whether plants can grow well. When the soil contains the right amount of water, air and nutrient, it becomes ideal for plants to grow, with roots spreading easily and getting the nutrients and water they

need; if the soil is too dry or too wet or lacking nutrients, plants grow with difficulty and may become weak or die. Testing shows whether the soil needs improving before planting, as farmers and gardeners often mix different types of soil or add fertilizers and organic matter to make soil fertile.

(ii) How can human activities cause soil pollution?

Human activities cause soil pollution in several ways: acid rain mixes pollutants from the air with the soil and changes its texture; agricultural activities contribute by using too many fertilizers and pesticides in farming; industrial activities contribute to soil pollution; dangerous chemicals and improper disposal of waste cause soil pollution; oil and fuels dumped into the soil harm it, especially the top of the soil; and deforestation, cutting down large numbers of trees, leaves the soil directly exposed to wind, rain and sunlight, making it weak and easy to wash or blow away.

(iii) A farmer has soil that is too sandy. How can he improve it for better plant growth?

A farmer with too-sandy soil can improve it the way farmers and gardeners improve poor soil before planting: by mixing different types of soil together, or by adding fertilizers and organic matter to make the soil fertile. Sandy soil is poor for growing plants because of its poor water holding capacity and low nutrients, and adding organic matter and fertilizer gives it the nutrients and water-holding ability it lacks, so plants grow faster, healthier and stronger.

(iv) Name the different types of soil.

The different types of soil in the unit are sandy soil, silty soil, clayey soil and loamy soil. They differ in their mixture of sand, clay, silt particles and organic matter: sandy soil is rough and light brown, silty soil is soft and smooth and brown or dark brown, clayey soil is fine, smooth but sticky and grey or dark grey, and loamy soil is soft and crumbly and dark brown or black.

(v) What are the professions related to Earth science?

The professions related to Earth science in the unit are palaeontologists, seismologists and geologists. Palaeontologists study fossils, including animal and plant bodies or their imprints, which are proof of life forms on Earth, and help us learn about dinosaurs and the history of Earth. Seismologists study how earthquakes happen, investigating the origin and movement of shock activity within the Earth and measuring earthquake strength. Geologists study the Earth, its physical structure, composition of layers, rocks and minerals, and help find natural resources such as water, oil, gas and minerals.

9.5 Write detailed answers of the following questions.

(i) Describe the different types of soil in detail.

Sandy soil is usually found near the seas or in deserts. It consists of small particles of weathered rock, is usually light brown, and its sand particles are large enough to be seen distinctly; it is one of the poorest types of soil for growing plants because of poor water holding capacity and low nutrients, and plant roots cannot be held by it. Silty soil is mostly found near rivers and flood plains; it is usually brown or dark brown, has smaller particles compared to

sandy soil, holds more water than sandy soil, and is more fertile compared to other types. Clay soil is usually found in swamps; its colour is grey or dark brown, its particles are tightly packed together with no air spaces or very little space, and it has very good water storage quality, with the highest water content. Loamy soil is a combination of all types of soil, containing the beneficial properties of each: it contains nearly equal amounts of sand and silt with a smaller amount of clay, has good water holding capacity, and is more suitable for farming. The comparison table gives their composition as: sandy 80 to 100 percent sand, silty 70 to 80 percent silt, clayey 50 to 100 percent clay, and loamy 10 to 30 percent clay, 30 to 50 percent silt and 25 to 50 percent sand with high humus.

(ii) How can soil composition change, and how does it affect plant growth?

Different soils have different compositions, made of different amounts of sand, clay, silt and organic matter, and the main characteristics are texture, which depends on particle size, with sand having the largest particles and clay the smallest; colour, which depends on the organic matter and minerals present; pore spaces, where sandy soil has larger pore spaces than silty soil; and moisture, where sand has the lowest water content and clay the highest. Soil composition can change when farmers and gardeners mix different types of soil or add fertilizers and organic matter, and it also changes through pollution, acid rain and deforestation. When the soil contains the right amount of water, air and nutrient, it becomes ideal for plants: good soil lets roots spread easily and

gives them the nutrients and water they need. If soil is too dry or too wet or lacking nutrients, it becomes difficult for plants to grow, roots may not get enough air or water, and plants may become weak or even die.

9.6 Constructed response questions.

(i) How does soil support life on Earth?

Soil supports life on Earth because plants grow in soil, and it provides them minerals, air and water, which they need to grow, and it protects plant roots from the heat of the Sun. Many animals such as insects, worms and other small creatures live in the soil, and it provides shelter to these animals. Tiny living things called microorganisms are found in the soil, and they break down the remains of dead plants and animals, which keeps the soil fertile. Soil is made of mineral particles (45 percent), air (25 percent), water (25 percent) and organic matter (5 percent), which together support plant growth and life on Earth.

(ii) Write at least three ways in which soil is important for humans, animals and plants.

First, for plants: plants grow in soil, which provides them minerals, air and water, protects their roots from the heat of the Sun, and gives roots room to spread and take in nutrients. Second, for animals: many animals such as insects, worms and other small creatures live in the soil, and it provides them shelter, while microorganisms in the soil break down dead remains and keep the soil fertile. Third, for humans: soil grows the crops and food we eat, since farmers grow crops mostly in loamy soil because it is rich and

fertile, and geologists help us find important natural resources in the Earth, while healthy soil, with its minerals and organic matter, supports farming and life itself.

9.7 PROJECT

Project: Plant seeds in sand, clay and garden soil and observe their growth after a week. Teacher's note: Shows which soil is best for plants. This is a hands-on classroom task, not a written answer.

Unit 10: Space and Satellites

Unit 10: Space and Satellites. Space is one of the great regions that lies beyond the Earth and its atmosphere, which obviously does not have any air and is extremely vast. A satellite is an object, natural or artificial, that orbits a larger object in space, and artificial satellites help us with weather monitoring, the internet, telephone calls and GPS services.

10.1 Multiple choice questions.

(i) Satellite orbits larger object in:

- Ocean
- Desert
- ✓ Space
- Mountain

(ii) Moon is example of:

- Spaceship
- ✓ Natural satellite
- Galaxy
- Star

(iii) Weather satellite helps in:

- Painting pictures
- Fixing roads
- ✓ Predicting weather
- Cleaning air

(iv) GPS helps in:

- Growing plants
- ✓ Finding locations
- Watching movies
- Repairing machines

(v) NASA is space agency of:

- Pakistan
- China
- ✓ USA
- Russia

(vi) Communication satellites used for:

- Cooking food
- ✓ Sending signals
- Building houses
- Growing trees

(vii) SUPARCO was formed in:

- 2001
- 1985
- ✓ 1961
- 1973

10.2 Fill in the blanks.

(i) A ____ is natural/artificial object orbiting a planet.

satellite

(ii) The ____ was first artificial satellite launched by Soviet Union in 1957.

Sputnik 1

(iii) Communication satellites help transmit ____, ____, and internet signals.

TV, radio

(iv) Weather satellites provide information about ____ patterns and natural disasters.

weather

(v) Path followed by satellite around Earth is called an ____.

orbit

10.3 Match column A with column B.

Match: Communication satellites, Weather satellites, Navigation satellites, Astronomical satellites, Sputnik 1.

Communication satellites: Used to transmit TV, radio and internet signals. Weather satellites: Provide information on storms and clouds. Navigation satellites: Helps in GPS and navigation. Astronomical satellites: Study stars, planets and galaxies. Sputnik 1: First artificial satellite launched in 1957.

10.4 Write short answers of the following questions.

(i) Define natural satellite and give one example.

A natural satellite is a celestial body that is not made by humans; it

revolves around a planet on its own accord, because of gravity. The best example is the Moon, which continuously orbits planet Earth; Mars has two small moons, Jupiter has more than 90 moons, and Saturn also has many moons.

(ii) Write one use of a communication satellite.

One use of a communication satellite is to send and receive signals for things like TV, cell phones and the internet. Communication satellites have special antennas and transponders that receive signals and send them back to Earth, and because many of them stay in a fixed position above the Earth, they give continuous service to specific areas, like a region or country, helping with live TV broadcasts and long distance phone calls, even in remote areas.

(iii) Name the first artificial satellite.

The first artificial satellite was Sputnik 1, launched by the Soviet Union in 1957. Space exploration began in 1957 with its launch, which marked the start of the space race.

(iv) What is the space agency of Pakistan?

The space agency of Pakistan is SUPARCO, the Pakistan Space and Upper Atmosphere Research Commission. It was established in 1961 and plays a crucial role in space exploration by conducting research, developing space technology and promoting its applications for national development and socio-economic benefit.

(v) Name two professions related to Earth science.

Two professions related to Earth science are astronauts and

physicists, along with space scientists. Astronauts are space explorers who travel to space and visit the Moon, stars and planets. Physicists study the laws of nature, like how things move, light, energy and forces, and space physicists study how these laws work in space. Space scientists study planets, stars and galaxies, using satellites and telescopes to discover new things about the universe.

10.5 Write detailed answers of the following questions.

(i) Describe the uses of artificial satellites in daily life.

Artificial satellites are machines created by humans, launched and placed in orbit around the Earth, and they are extremely helpful in numerous ways. Weather satellites, orbiting usually 500 to 22,000 miles above the surface, gather information about the weather using advanced cameras and sensors, detecting clouds, storms, temperature, humidity and wind speed, and are especially useful for spotting dangerous weather events like hurricanes, tornadoes and storms, giving early warnings. Communication satellites, usually about 22,000 miles above the surface, send and receive signals for TV, cell phones and the internet, staying in a fixed position to give continuous service. Navigation satellites form the global positioning system, a network of at least 24 satellites sending signals to GPS devices in phones, cars and ships, so the device can figure out exactly where you are and give directions. Military satellites are used for surveillance, intelligence gathering and military navigation. Astronomical satellites study stars, planets and galaxies.

(ii) Describe the role of SUPARCO in space exploration.

SUPARCO, the Pakistan Space and Upper Atmosphere Research Commission, plays a crucial role in space exploration by conducting research, developing space technology and promoting its applications for national development and socio-economic benefit. It was established in 1961. Its responsibilities include satellite design and development, space science research and the establishment of infrastructure for related activities. Pakistan sent its first artificial satellite, Badar 1, into space on 16th July 1990, and SUPARCO is, with NASA of the USA, one of the two major space agencies named in the unit that explore space and develop new space technologies.

(iii) Write a short note on GPS and how it works.

The global positioning system (GPS) is a network of satellites that help us know where we are on Earth; it works like a giant map in the sky. There are at least 24 satellites orbiting the Earth that send signals to GPS devices on the ground, like in phones, cars and even ships. When you use GPS, the device receives signals from several satellites, and by measuring how long it takes for the signals to travel, the GPS can figure out exactly where you are and give directions, whether you are walking, driving or even flying. GPS is super useful for navigation: it helps people find their way to places, track routes and figure out how far away something is, and it is also used in science and emergency services, like helping rescuers find people who are in need. There are only 24 satellites operational in the GPS system, with a backup of three more satellites to replace a failed one.

10.6 Constructed response question.

How do artificial satellites help people in their daily lives? Write at least three ways satellites are used and how they make life easier.

Artificial satellites help people in daily life in at least three ways. First, communication: communication satellites send and receive signals for TV, cell phones and the internet, staying in a fixed position above the Earth to give continuous service, so people can watch live TV broadcasts, make long distance phone calls and stay connected even in remote areas. Second, weather: weather satellites track storms, wind and clouds to help scientists predict the weather and give early warnings, especially for dangerous events like hurricanes, tornadoes and storms. Third, navigation: GPS satellites help us find our exact location and give directions through phones, cars and other devices, so we can find our way, track routes and figure out distances, and they also help rescuers find people in need. Together, these uses make life easier by keeping us connected, warning us of danger and guiding us wherever we go.

10.7 PROJECT

Project: Use a flashlight as the Sun, a globe as the Earth and a ball as the Moon, and show how shadows create eclipses. Teacher's note: Demonstrates the alignment of the Sun, Earth and Moon during eclipses. This is a hands-on classroom task, not a written answer.

Notes, Flags and Print Corrections

This chapter records every place where the printed book is unclear, misprinted or incomplete, and how these notes handle it. Nothing here changes any exercise answer; it only explains the choices openly.

1.2 Fill-in numbering: In Unit 1 the printed exercise numbers the fill-in items i, ii, iii, vi, v, with no item iv in print. These notes preserve the book's printed numbering, with item vi before item v, exactly as printed.

2.3 Match, Alveoli item: The Unit 2 match exercise pairs Alveoli with 'Tiny air sacs for gas exchange in humans'. That is Unit 1 content appearing in the Unit 2 exercise; it is reproduced as printed.

2.3 Match, year 1683: The match pairs Antonie Van Leeuwenhoek with 'First observed microorganisms in 1683'. The lesson (p.24) confirms he first observed microorganisms on September 17, 1683, so the printed year stands.

3.6 Classify, Fish: The unit lists fish only as an example of prey (p.44) and never states its diet. These notes place Fish in Omnivores as the standard school classification, and say so openly here; Deer is confirmed as a herbivore and primary consumer (p.46), Wolf as a predator (p.44) and Bear as an omnivore (p.44).

4.1(v) Burning or rusting: The answer Burning follows the Key Points definition of burning as a chemical reaction of a substance with oxygen to produce heat or light. Rusting is specifically the reaction of iron with oxygen in the presence of moisture, so it is the

narrower term and is not used for this general question.

4.6(i) Iron with no moisture: The answer applies the book's own definition of rusting (iron plus oxygen plus moisture, p.74), so without moisture rusting does not occur. This is an application of the book's stated rule, not a new fact.

5.1(i) Option letters misprinted: The printed options for this question are lettered a, b, d, d, with no option c. These notes use the correct option positions, with the Sun as the second option, and preserve the book's option order.

6.2(iii) Power source: The battery is confirmed as a power source by the lesson (p.95), which lists Power source as the first component of an electric circuit and says it may be a battery or an electric cell.

7.1(v) Fire and smoke: The answer 'Lower yourself' is verbatim from the lesson (p.113): 'If there is smoke, lower yourself to the ground and move on.'

7.2(iv) Level: The lesson (pp.110-111) says that if the bubble comes exactly in the center, the surface is perfectly flat, and the activity (p.111) asks whether the bubble shows the surface is level or not. The fill word 'level' is therefore the book's own term for a perfectly flat surface.

8.2(iv) Mantle depth: The lesson (p.119) says the mantle is approximately 2900 km thick, and the outer core starts about 2890 km beneath the surface. The fill answer 2900 km uses the book's own mantle figure.

8.2(v) Outer core composition: The lesson (p.120) and Figure 8.1 both give the outer core as a layer of liquid iron (Fe) and nickel (Ni), so the answer is iron and nickel.

8.4(iii) and 8.6(i) Why the interior is hot: The unit states the

temperatures of each layer but never explains the cause of the heat. These notes therefore describe the book's own temperature pattern, from about 22 degrees Celsius at the crust to between 5000 and 6000 degrees Celsius at the inner core, rather than inventing a cause the book does not give.

9.1(v) Geologist: The multiple choice answer follows the question's own option, 'Earth layer'. The unit's match pairs Geologists with 'Studies rocks', and the lesson (p.134) says geologists study the Earth, its physical structure, composition of layers, rocks and minerals, so both wordings come from the book itself.

9.1(vii), 9.2(iv) and 9.2(v) are Unit 8 questions inside the Unit 9 exercise: The inner core is about 2500 km wide and its temperature ranges between 5000 and 6000 degrees Celsius (Unit 8 lesson, p.120), and molten rock beneath the Earth's surface is called magma (Unit 8 lesson, p.120). They are answered from the Unit 8 lesson text, not from general knowledge.

9.2(ii) Clayey soil: The exercise leaves this blank open-ended ('Clayey soil is ____.'). The answer 'fine and sticky' uses the book's own comparison table (Table 9.1, p.128), which gives clayey soil texture as 'Fine (smooth but sticky)'; the lesson also says clay particles are tightly packed with no air spaces and have very good water storage.

9.2(iv) Garbled sentence: The printed sentence reads 'The method rock beneath the Earth's surface is called ____.' The word 'method' is a misprint for 'molten'; the answer is magma, the book's term for melted rock while it is still underground (Unit 8, p.120).

10.1(vii) SUPARCO year: The lesson (p.142) states SUPARCO was established in 1961, so the answer is 1961.

Glossary (pp.148-152) and Index (pp.153-154): Both are word lists, definitions and page references, with no exercises, so they carry no solved items in these notes, exactly as the Glossary and Index of the Grade 4 book were handled. Note that the contents page lists the Index at p.151, but pp.151-152 are in fact the end of the Glossary; the Index itself starts on p.153.

Projects: Every X.7 PROJECT in the book is a hands-on classroom task with a short guide in these notes, not a solved question, because a project has no single written answer.

Match exercises: Each match exercise is given as one solved item listing all pairs, because the builder has no separate match item kind. Sort (2.6) and Classify (3.6) tasks are solved as listed groupings, and Constructed response questions are answered as medium-length items grounded in lesson text.